Boletín 0. Números complejos

① Calcular

(a) $(1+3i)+(2-2i) = 3+i$ b) $(5-4i)-(6-7i) = -1+3i$

c) $(6+8i)(1-i) = 6-6i+8i-8\overset{-1}{i^2} = 6+2i+8 = 14+2i$

d) $(2+i)^3 = (2^2+2i+2i+i^2)(2+i) = (3+4i)(2+i) = 6+3i+8i-4 = 2+11i$

e) $i^{343} = (i^4)^{85}\cdot i^3 = i^3 = -i$

f) $(-2i)^{-1} = \frac{-1}{2i} = \frac{-1}{2i}\,\frac{i}{i} = \frac{i}{2} = \frac{1}{2}i$

g) $\frac{4+5i}{3+4i} = \frac{4+5i}{3+4i}\,\frac{3-4i}{3-4i} = \frac{12-16i+15i-20i^2}{9+16} = \frac{32-i}{25} = \frac{32}{25}-\frac{1}{25}i$

h) $\frac{i^7(3-2i)}{(2-i)i} = \frac{-i(3-2i)}{1+2i} = \frac{-2-3i}{1+2i}\,\frac{1-2i}{1-2i} = \frac{-2+4i-3i-6}{1+4} = \frac{-8}{5}+\frac{1}{5}i$

i) $\frac{3(1-2i)+(2+i)^2}{(1+i)(1-3i)-4} = \frac{3-6i+4-1+4i}{1-3i+i+3-4} = \frac{6-2i}{-2i} = \frac{-3+i}{i} = \frac{-1-3i}{-1} = 1+3i$

② Calcular el conjugado de cada uno de los resultados del ejercicio anterior.

a) $z = 3+i \Rightarrow \bar{z} = 3-i$

b) $z = -1+3i \Rightarrow \bar{z} = -1-3i$

c) $z = 14+2i \Rightarrow \bar{z} = 14-2i$

d) $z = 2+11i \Rightarrow \bar{z} = 2-11i$

e) $z = 0-i \Rightarrow \bar{z} = 0+i$

f) $z = 0+\frac{1}{2}i \Rightarrow \bar{z} = 0-\frac{1}{2}i$

g) $z = \frac{32}{25}-\frac{1}{25}i \Rightarrow \bar{z} = \frac{32}{25}+\frac{1}{25}i$

h) $z = -\frac{8}{5}+\frac{1}{5}i \Rightarrow \bar{z} = -\frac{8}{5}-\frac{1}{5}i$

i) $z = 1+3i \Rightarrow \bar{z} = 1-3i$

③ Calcular el módulo y el argumento de los siguientes números complejos:

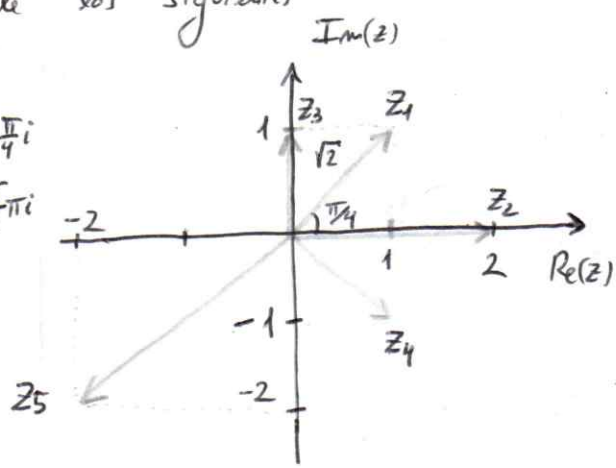
$z_1 = 1+i = \sqrt{2}e^{i\frac{\pi}{4}}$

$z_2 = 2 = 2e^{0i}$

$z_3 = i = 1\cdot e^{i\frac{\pi}{2}}$

$z_4 = 1-i = \sqrt{2}e^{-\frac{\pi}{4}i}$

$z_5 = -2-2i = 2\sqrt{2}e^{\frac{5}{4}\pi i}$



④ Hallar dos números complejos tales que su suma sea $(1+4i)$, su cociente sea imaginario puro y la parte real de uno de ellos sea -1. $z_1 = a+bi$, $w = c+di$.

$a+bi + c+di = 1+4i$

$Re\left(\frac{a+bi}{c+di}\right) = 0$

$a = -1$

Si $a=-1 \Rightarrow c=2$. Además $b+d=4$

$$\frac{a+bi}{c+di} = \frac{a+bi}{c+di}\,\frac{c-di}{c-di} = \frac{ac-adi+cbi+bd}{c^2+d^2} = \frac{ac+bd+(cb-ad)i}{c^2+d^2}$$

$\Rightarrow ac+bd=0 \Rightarrow b(4-b) = +2 \Rightarrow$

$\Rightarrow b^2-4b+2=0 \Rightarrow b = \frac{4\pm\sqrt{16-4\cdot 2}}{2\cdot 1} = \frac{4\pm\sqrt{8}}{2} = 2\pm\sqrt{2}$

$\Rightarrow$ Dos soluciones: $\begin{cases} z = -1+(2+\sqrt{2})i \\ w = 2+(2-\sqrt{2})i \end{cases}$ $\begin{cases} z = -1+(2-\sqrt{2})i \\ w = 2+(2+\sqrt{2})i \end{cases}$

(Nota: dado que en principio no es equivalente tomar la condición $a=-1$ o $c=-1$ en $Re\left(\frac{a+bi}{c+di}\right)=0$, uno podría pensar que hay más soluciones, pero al desarrollar dicha condición se obtiene que sí es equivalente.)

⑤ Hallar dos números complejos z y w tales que $z+w$ sea imaginario puro, $z-w=2$ y $z\cdot w=-2$.

$z = a+bi \quad w = c+di$

$z+w = a+c+(b+d)i \qquad Re(z+w)=0 \Rightarrow a+c=0 \Rightarrow a=-c$

$z-w = a-c+(b-d)i = -2 \Rightarrow a-c=2$ y $b-d=0 \Rightarrow a=1,\ b=d$

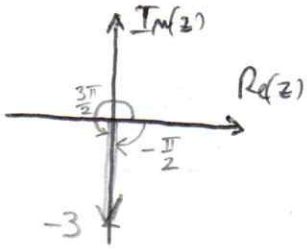
$z\cdot w = (1+bi)(-1+bi) = -1+bi-bi-b^2 = -1-b^2 = -2 \Rightarrow b^2=1 \Rightarrow b=\pm 1$

Dos soluciones: $\begin{cases} z = 1+i \\ w = -1+i \end{cases}$ $\begin{cases} z = 1-i \\ w = -1-i \end{cases}$

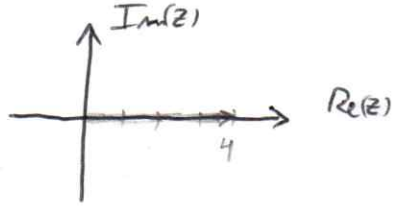
⑥ Expresar cada uno de los siguientes números complejos en sus formas trigonométricas, binómica, exponencial y polar:

a) $-3i = 3e^{-\frac{\pi}{2}i} = 3\left(\overbrace{\cos\frac{\pi}{2}}^{0} - \overbrace{\sin\frac{\pi}{2}}^{1} i\right) = 3_{-\frac{\pi}{2}}$

BINÓMICA EXPONENCIAL TRIGONOMÉTRICA POLAR

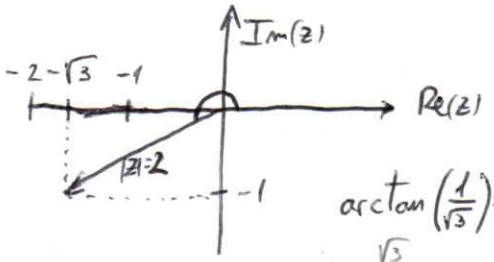


b) $4 = 4e^{0i} = 4(\cos 0 + \sin 0\, i) = 4_0$

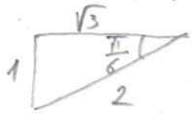


c) $\boxed{-\sqrt{3} - i = 2e^{\frac{7}{6}\pi i} = 2\left(\cos\frac{7\pi}{6} + \sin\frac{7\pi}{6}\, i\right) = 2_{\frac{7\pi}{6}}}$

BINÓMICA EXPONENCIAL TRIGONOMÉTRICA POLAR



$\arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

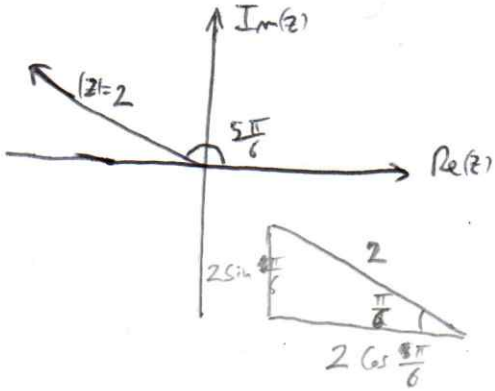


d) $\sqrt{8}\left(\overbrace{\cos\pi}^{-1} + i\,\overbrace{\operatorname{sen}\pi}^{0}\right) = \sqrt{8}\, e^{i\pi}$

$\boxed{-\sqrt{8} + 0i = \sqrt{8}\, e^{i\pi} = \sqrt{8}\,(\cos\pi + i\operatorname{sen}\pi) = \sqrt{8}_{\pi}}$

e) $2e^{\frac{5\pi}{6}i} = 2\left(\underbrace{\cos\frac{5\pi}{6}}_{-\sqrt{3}/2} + i\,\underbrace{\sin\frac{5\pi}{6}}_{+1/2}\right) = -\sqrt{3} + i$

$\boxed{-\sqrt{3} + i = 2e^{\frac{5\pi}{6}i} = 2\left(\cos\frac{5\pi}{6} + \sin\frac{5\pi}{6}\, i\right) = 2_{\frac{5\pi}{6}}}$



f) $5e^{i\frac{\pi}{3}} = 5\left(\overbrace{\cos\frac{\pi}{3}}^{1/2} + \overbrace{\sin\frac{\pi}{3}}^{\sqrt{3}/2}\, i\right) = \frac{5}{2} + \frac{5\sqrt{3}}{2}i$

$\boxed{\frac{5}{2} + \frac{5\sqrt{3}}{2}i = 5e^{i\frac{\pi}{3}} = 5\left(\cos\frac{\pi}{3} + \sin\frac{\pi}{3}\, i\right) = 5_{\frac{\pi}{3}}}$

g) $8\left(\underbrace{\cos\frac{\pi}{4}}_{\sqrt{2}/2} + i\,\underbrace{\operatorname{sen}\frac{\pi}{4}}_{\sqrt{2}/2}\right) = 4\sqrt{2} + 4\sqrt{2}i$

$\boxed{4\sqrt{2} + 4\sqrt{2}\, i = 8e^{\frac{\pi}{4}i} = 8\left(\cos\frac{\pi}{4} + \operatorname{sen}\frac{\pi}{4}\, i\right) = 8_{\frac{\pi}{4}}}$

h) $9_{\pi/4} = 9e^{\frac{\pi}{4}i} = 9\left(\cos\frac{\pi}{4} + \sin\frac{\pi}{4}\, i\right) = \frac{9\sqrt{2}}{2} + \frac{9\sqrt{2}}{2}i$

$\boxed{\frac{9\sqrt{2}}{2} + \frac{9\sqrt{2}}{2}i = 9e^{\frac{\pi}{4}i} = 9\left(\cos\frac{\pi}{4} + \sin\frac{\pi}{4}\, i\right) = 9_{\pi/4}}$

i) $3_{\frac{5\pi}{2}} = 3e^{\frac{5\pi}{2}i} = 3e^{\frac{\pi}{2}i} = 3\left(\cos\frac{\pi}{2} + \sin\frac{\pi}{2}\, i\right) = 3i$

$\boxed{0 + 3i = 3e^{\frac{5\pi}{2}i} = 3\left(\cos\frac{\pi}{2} + \sin\frac{\pi}{2}\, i\right) = 3_{\frac{\pi}{2}}}$

(7) Sean $z, w_1, w_2 \in \mathbb{C}$ números complejos de módulo uno.

a) Demostrar que $z^{-1} = \bar{z}$.

$$z^{-1} = \frac{1}{z} = \frac{\bar{z}}{z\cdot\bar{z}} = \frac{\bar{z}}{|z|^2} = \bar{z}$$

$|z| = 1$

b) Demostrar que $w_1\overline{w_2} + \overline{w_1}w_2 + \frac{1}{w_1\overline{w_2}} + \frac{1}{\overline{w_1}w_2} = 4\,Re(w_1\overline{w_2})$

Usando el apartado a): $\frac{1}{w_1} = \overline{w_1}$, $\frac{1}{\overline{w_2}} = w_2$, etc:

$$w_1\overline{w_2} + \overline{w_1}w_2 + \frac{1}{w_1\overline{w_2}} + \frac{1}{\overline{w_1}w_2} = 2w_1\overline{w_2} + 2\overline{w_1}w_2 = 2(w_1\overline{w_2} + \overline{w_1\overline{w_2}}) = 4Re(w_1\overline{w_2})$$

$z = a+bi \Rightarrow z+\overline{z} = 2a = 2Re(z)$

(8) Hallar $z \in \mathbb{C}$ para que $\frac{3+zi}{1-i}$ sea un número real y $(4+i)z$ sea un número imaginario puro.

Sea $z = a+bi$. $\quad \frac{3+ai-b}{1-i} = \frac{(3-b+ai)(1+i)}{1+1} = \frac{1}{2}(3-b+ai+(3-b)i-a)$

Entonces: $Im\left(\frac{3+zi}{1-i}\right) = 0 \Rightarrow a-b+3=0$

$(4+i)z = (4+i)(a+bi) = 4a+4bi+ai-b = (4a-b)+(4b+a)i$

$Re((4+i)z) = 0 \Rightarrow 4a-b=0 \Rightarrow b=4a$

$\left.\begin{array}{l} b=4a \\ a-b+3=0 \end{array}\right| \Rightarrow -3a+3=0 \Rightarrow a=1 \Rightarrow b=4$. $\boxed{\text{Solución: } z = 1+4i}$

(9) Expresar en forma binómica los números complejos z tales que $z^4 = -3$.

Lo pasamos a forma exponencial: $w = -3+0i = 3e^{i\pi}$

Usamos la fórmula para obtener las raíces:

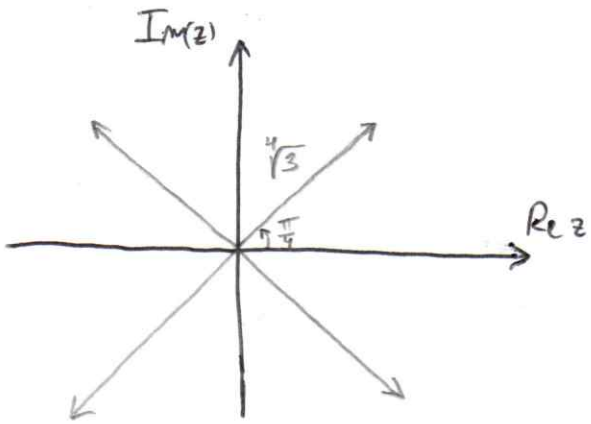
$$z_K = |w|^{1/4} e^{i\left(\frac{\alpha+2\pi K}{4}\right)} \qquad K=0,1,2,3$$

$\sqrt[4]{3}\, e^{\frac{\pi}{4}i} = z_0 = \sqrt[4]{3}\frac{\sqrt{2}}{2}(1+i)$

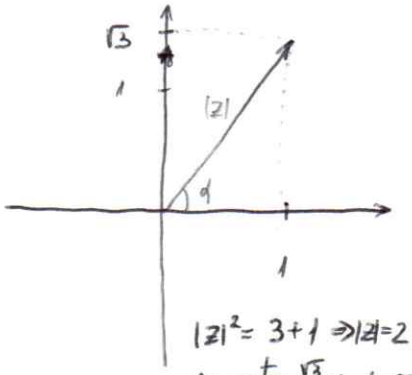
$\sqrt[4]{3}\, e^{\frac{3\pi}{4}i} = z_1 = \sqrt[4]{3}\frac{\sqrt{2}}{2}(-1+i)$

$\sqrt[4]{3}\, e^{\frac{5\pi}{4}i} = z_2 = \sqrt[4]{3}\frac{\sqrt{2}}{2}(-1-i)$

$\sqrt[4]{3}\, e^{\frac{7\pi}{4}i} = z_3 = \sqrt[4]{3}\frac{\sqrt{2}}{2}(+1-i)$



(10) Calcular $(1+\sqrt{3}i)^8$



$|z|^2 = 3+1 \Rightarrow |z| = 2$

$\alpha = \arctan \frac{\sqrt{3}}{1} \Rightarrow \alpha = \frac{\pi}{3}$

Para calcular potencias, pasamos a forma exponencial. Ayudándonos del plano complejo y haciendo trigonometría, obtenemos:

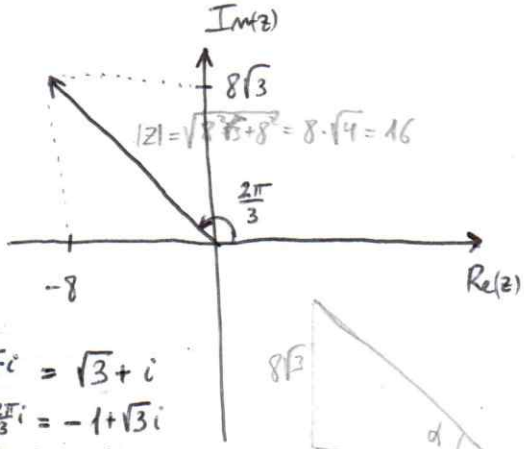
$$1+\sqrt{3}i = 2e^{i\frac{\pi}{3}}$$

Entonces: $\boxed{(1+\sqrt{3}i)^8} = \left(2e^{i\frac{\pi}{3}}\right)^8 = 2^8 e^{i\frac{8\pi}{3}} = \boxed{2^8 e^{\frac{2\pi}{3}i}}$

Si lo queremos poner en forma binómica:

$$2^8 e^{\frac{2\pi}{3}i} = 2^8\left(\cos\frac{2\pi}{3} + \sin\frac{2\pi}{3}i\right) = 2^7(-1+\sqrt{3}i)$$

(11) Sea $z = -8+8\sqrt{3}i$. Calcular $\sqrt[4]{z}$.



Para calcular raíces, pasamos a forma exponencial. Dibujando en el plano complejo, geométricamente podemos obtener: $z = -8+8\sqrt{3}i = 16e^{\frac{2\pi}{3}i}$

Las raíces cuartas de z son:

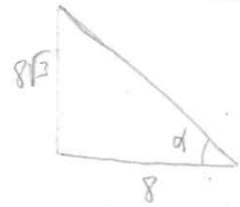
$$W_K = \sqrt[4]{16}\cdot e^{\left(\frac{\frac{2\pi}{3}+2\pi K}{4}\right)i} \qquad K=0,1,2,3$$

$W_0 = 2e^{\frac{\pi}{6}i} = \sqrt{3}+i$

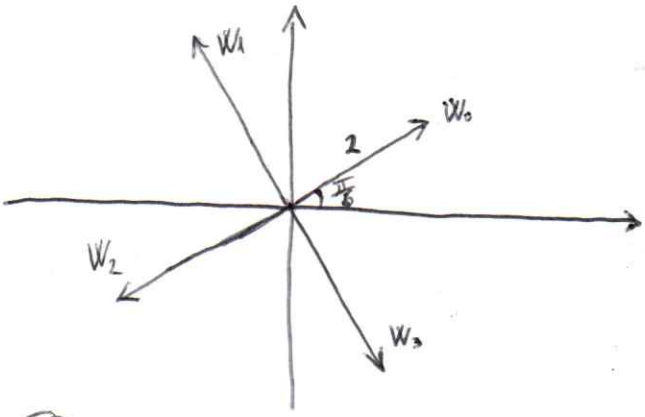
$W_1 = 2e^{\frac{2\pi}{3}i} = -1+\sqrt{3}i$

$W_2 = 2e^{\frac{7}{6}\pi i} = -\sqrt{3}-i$

$W_3 = 2e^{\frac{5}{3}\pi i} = 1-\sqrt{3}i$



$\alpha = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$



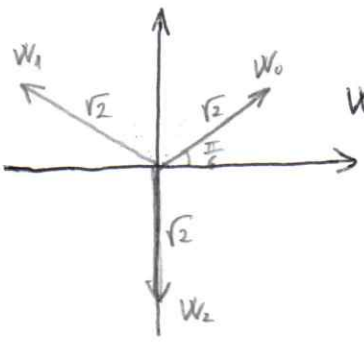
Podemos pasarlos a forma binómica:

$$W_0 = 2e^{\frac{\pi}{6}i} = 2\left(\underbrace{\cos\frac{\pi}{6}}_{\sqrt{3}/2} + \underbrace{\sin\frac{\pi}{6}}_{1/2}i\right) = \sqrt{3}+i$$

(12) Sea $z = \dfrac{-4+4\sqrt{3}i}{\sqrt{6}+\sqrt{2}i}$. Calcular las raíces cúbicas de z.

Para ello pasamos a forma exponencial. Necesitamos primero simplificar la expresión:

$$z = \frac{-4+4\sqrt{3}i}{\sqrt{6}+\sqrt{2}i} = \frac{4}{\sqrt{2}}\,\frac{-1+\sqrt{3}i}{\sqrt{3}+i}\,\frac{\sqrt{3}-i}{\sqrt{3}-i} = \frac{4}{\sqrt{2}}\,\frac{-\sqrt{3}+i+3i+\sqrt{3}}{3+1} = 2\sqrt{2}i = 2\sqrt{2}e^{i\frac{\pi}{2}}$$



$$W_K = |z|^{1/3} e^{\left(\frac{\alpha+2\pi K}{3}\right)} = \sqrt{2}\,e^{i\left(\frac{\pi}{6}+\frac{2}{3}\pi K\right)}$$

$W_0 = \sqrt{2}e^{\frac{\pi}{6}i} = \frac{\sqrt{6}}{2}+\frac{\sqrt{2}}{2}i$

$W_1 = \sqrt{2}e^{\frac{5\pi}{6}i} = -\frac{\sqrt{6}}{2}+\frac{\sqrt{2}}{2}i$

$W_2 = \sqrt{2}e^{\frac{9\pi}{6}i} = -\sqrt{2}i$

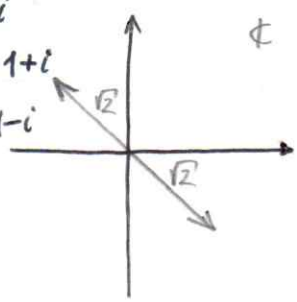
(13) Resolver las siguientes ecuaciones:

a) $z^2-(1+i)z+i=0$

Usamos la fórmula general: $z=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}=\dfrac{1+i\pm\sqrt{(1+i)^2-4i}}{2}$

Simplificamos el discriminante: $\Delta=(1+i)^2-4i=1-1+2i-4i=-2i$

Calculamos la raíz: $\sqrt{-2i}=\sqrt{2e^{+\frac{3\pi}{2}i}}=\sqrt{2}\,e^{i\left(\frac{3\pi}{4}+\pi k\right)}=\begin{cases}\sqrt{2}\,e^{i\frac{3\pi}{4}}=-1+i\\ \sqrt{2}\,e^{i\frac{7\pi}{4}}=1-i\end{cases}$



Entonces las dos soluciones de la ecuación son:

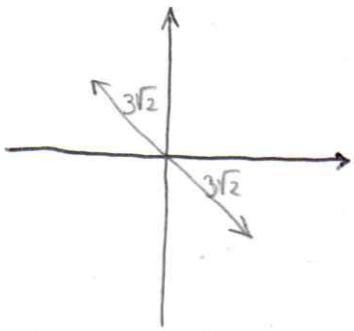
$$z=\frac{1+i\pm(-1+i)}{2}=\begin{cases}\frac{1}{2}2i=i\\ \frac{1}{2}2=1\end{cases}$$

$\boxed{\text{Soluciones } z=1,\ z=i}$

b) $z^2-5(1+i)z+17i=0$

$z=\dfrac{5(1+i)\pm\sqrt{25(1+i)^2-4\cdot 17i}}{2}$ $\qquad \Delta=25(1-1+2i)-4\cdot 17i=50i-68i=-18i$

$\sqrt{-18i}=\sqrt{18e^{\frac{3\pi}{2}i}}=3\sqrt{2}\,e^{\left(\frac{3\pi}{4}+\pi k\right)i}=\begin{cases}3\sqrt{2}\,e^{\frac{3\pi}{4}i}=-3+3i\\ 3\sqrt{2}\,e^{\frac{7\pi}{4}i}=+3-3i\end{cases}$

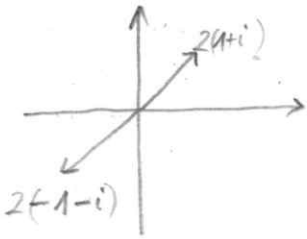


$z=\frac{1}{2}\left(5+5i\pm(3-3i)\right)=\begin{cases}\frac{1}{2}(8+2i)=4+i\\ \frac{1}{2}(2+8i)=1+4i\end{cases}$

c) $z^2i-2iz+i-2=0$

$z=\dfrac{2i\pm\sqrt{-4+4(-1+2i)}}{2i}=\dfrac{2i\pm\sqrt{8i}}{2i}$

$\sqrt{8i}=\sqrt{8e^{i\frac{\pi}{2}}}=\sqrt{8}\,e^{i\left(\frac{\frac{\pi}{2}+2\pi k}{2}\right)}=\begin{cases}2\sqrt{2}\,e^{i\frac{\pi}{4}}=2(1+i)\\ 2\sqrt{2}\,e^{i\frac{5\pi}{4}}=2(-1-i)\end{cases}$



$z=\dfrac{2i\pm 2(1+i)}{2i}=\begin{cases}\dfrac{2i+2+2i}{2i}=\dfrac{2+4i}{2i}=\dfrac{1+2i}{i}\dfrac{i}{i}=2-i\\ \dfrac{2i-2-2i}{2i}=\dfrac{-2}{2i}\dfrac{i}{i}=i\end{cases}$

d) $z^2 - 6z + 11 = 0$

$$z = \frac{6 \pm \sqrt{36 - 4 \cdot 11}}{2} = \frac{6 \pm \sqrt{-8}}{2} = 3 \pm \sqrt{-2}$$

$$\sqrt{-2} = \sqrt{2\, e^{\pi i}} = \sqrt{2}\; e^{i\left(\frac{\pi + 2\pi k}{2}\right)} = \begin{cases} \sqrt{2}\, e^{i\frac{\pi}{2}} = \sqrt{2}\, i \\ \sqrt{2}\, e^{i\frac{3\pi}{2}} = -\sqrt{2}\, i \end{cases}$$

Entonces:

$$z = 3 \pm \sqrt{-2} = \begin{cases} 3 + \sqrt{2}\, i \\ 3 - \sqrt{2}\, i \end{cases}$$

e) $\bar{z} = z^2$

Sea $z = a + bi \Rightarrow a - bi = a^2 - b^2 + 2abi \iff \begin{cases} a = a^2 - b^2 \\ -b = 2ab \end{cases}$

De la segunda ecuación: si $b \neq 0 \Rightarrow 2a = -1 \Rightarrow a = -1/2$

Reemplazando en la primera ecuación: $b^2 = a^2 - a = \frac{1}{4} + \frac{2}{4} = \frac{3}{4} \Rightarrow b = \frac{\sqrt{3}}{2}$

$\Rightarrow z = -\frac{1}{2} + \frac{\sqrt{3}}{2} i$

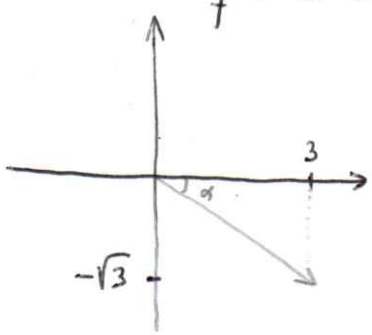
(14) Resuelve la ecuación $|e^{i\theta} - 1| = 2$ para $\theta \in [0, 2\pi)$

$$|\cos\theta + \operatorname{sen}\theta\, i - 1| = |\cos\theta - 1 + \operatorname{sen}\theta\, i| = \sqrt{(\cos\theta - 1)^2 + \sin^2\theta} =$$

$$= \sqrt{\cos^2\theta + 1 - 2\cos\theta + \sin^2\theta} = \sqrt{2 - 2\cos\theta} = 2 \Rightarrow 2 - 2\cos\theta = 4$$

$$1 - \cos\theta = 2 \Rightarrow \cos\theta = -1 \iff \theta = \pi$$

(15) Calcular el único número complejo $z \in \mathbb{C}$ cuyo argumento es el mismo que el de $3 - \sqrt{3}\, i$ y tal que $\operatorname{Re}(zi) = 3$.



$\alpha = \arctan \frac{\sqrt{3}}{3} = \frac{\pi}{6}$. El argumento es $-\frac{\pi}{6}$

$z = a + bi \qquad zi = -b + ai \Rightarrow R(zi) = -b = 3 \Rightarrow b = -3$

$|z|\, e^{-\frac{\pi}{6} i} = |z| \left(\cos\left(-\frac{\pi}{6}\right) + \operatorname{sen}\left(-\frac{\pi}{6}\right) i\right) \Rightarrow |z| \underbrace{\operatorname{sen}\left(-\frac{\pi}{6}\right)}_{-1/2} = -3$

$\Rightarrow |z| = 6 \Rightarrow z = 6\left(\cos\left(-\frac{\pi}{6}\right) + \operatorname{sen}\left(-\frac{\pi}{6}\right) i\right) = \boxed{3\sqrt{3} - 3i}$